

JAVA MINI PROJECT

Restaurant Order and Billing System

Project Type: Java Application

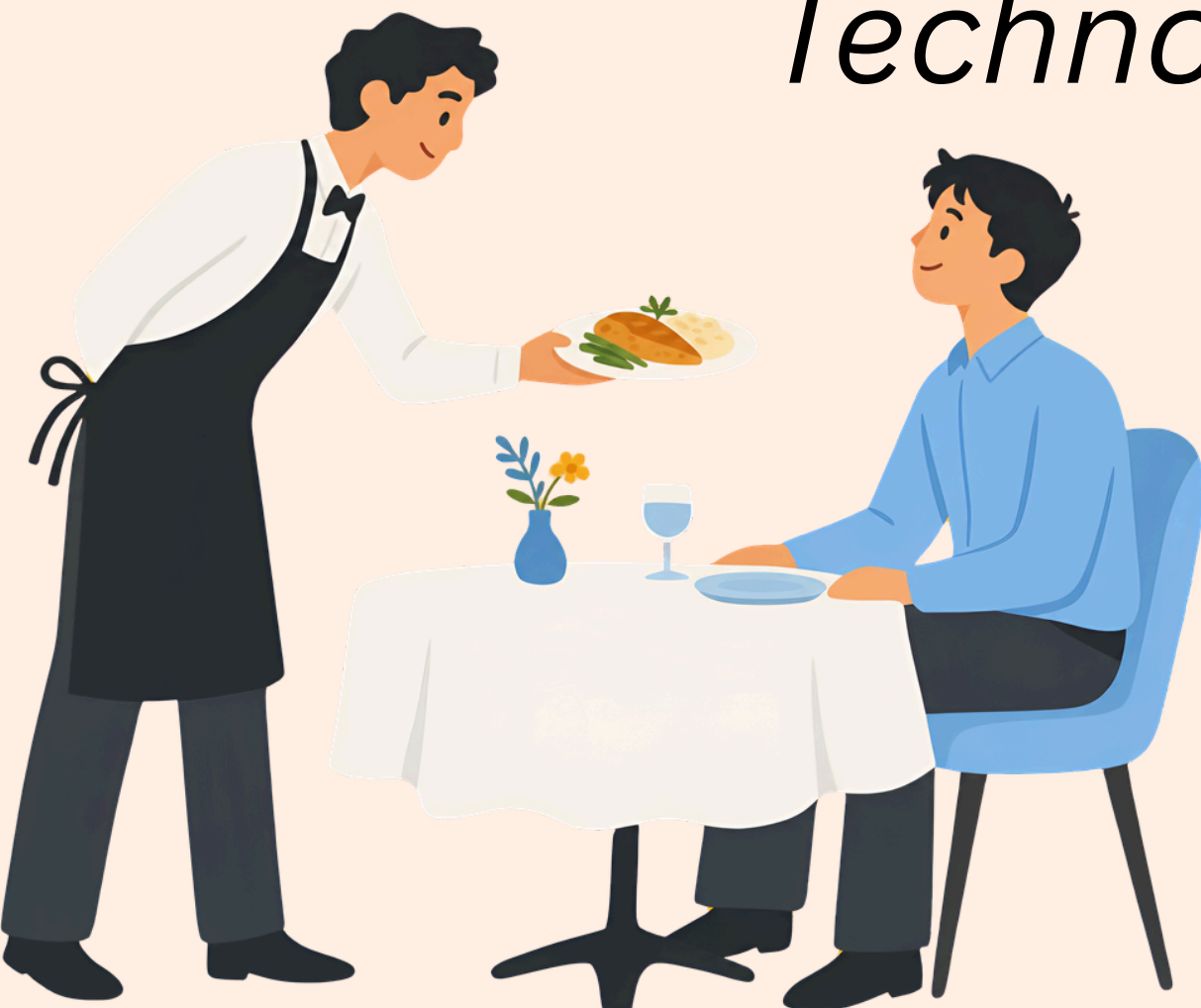
Technology: Java + Swing GUI

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< Problem Statement >

A restaurant needs a computerized system to manage menu items, customer orders, quantities, discounts, taxes, billing, and order status efficiently.

< Objectives >

- *Maintain & manage menu items*
- *Create & modify customer orders*
- *Calculate subtotal, discounts, taxes, & final bill*
- *Track order status*
- *Generate sales summaries*
- *Reduce manual calculation & record-keeping errors*

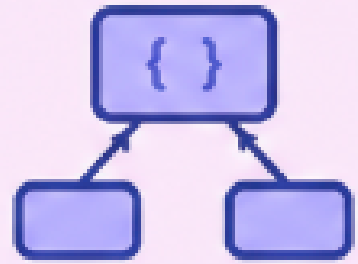


< Main Features >

- Menu Management
- Customer Order Management
- Quantity Management
- Automatic Bill Calculation
- Discount and Tax Calculation
- Order Status Tracking
- Sales Summary



Core Features Implemented



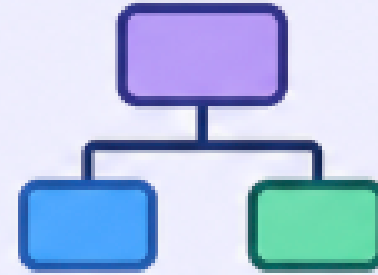
Classes and Objects

Blueprints and instances of a class.



Constructors

Initialize objects with values.



Inheritance

Reuse code from parent class.



Encapsulation

Keep data private and access via methods.



Exception Handling

Handle runtime errors gracefully.



BigDecimal

Accurate decimal calculations.



ArrayList / Collections

Store and manage multiple values.

Java Concepts Used



Swing GUI

Create graphical user interfaces.



LocalDate

Work with dates and time.

```
add(int, int)
```

```
add(double, double)
```

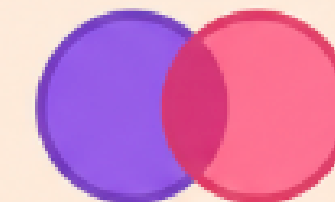
Method Overloading & Overriding

Multiple forms of a method and class-specific behaviour.



Streams

Process data using functional style.



Polymorphism

Same method, different behaviour.



File I/O

Read and write files.

System Design



Menu Management

Add, update, and manage menu items.



Sales Summary

Display completed orders and sales.



Order Management

Create and modify customer orders.

Item	Amount
Subtotal	₹500
Discount	₹50
Tax (5%)	₹22.50
Total	₹472.50

Billing System

Calculate subtotal, discount, tax, and total.



Order Status

Track order progress.

Bill Generation

Generate and save printable bills



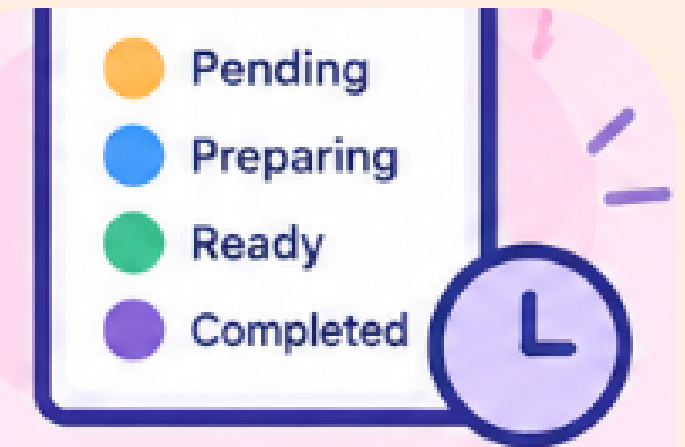
< Project Outcomes >



Successfully digitizes
restaurant order
management



Provides accurate and
automated bill calculation



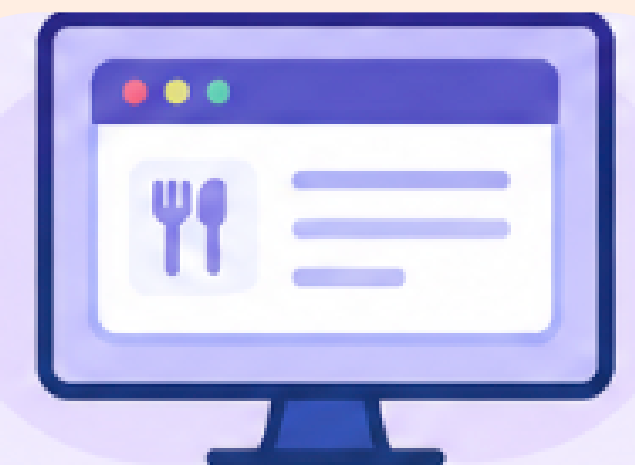
Supports order
status management



Reduces manual work
and calculation errors



Demonstrates important
Java programming
concepts



Provides a simple
and user-friendly GUI

< **Conclusion** >

The Restaurant Order and Billing System demonstrates how Java can be used to develop a practical application for restaurant operations while applying object-oriented programming, collections, exception handling, file handling, and GUI development.

Thank You!!

